Optional Self-Reflection: Unstructured data

Activity Overview

Now that you've learned about unstructured data and data formats, pause for a moment and think about unstructured data and how it compares to structured data. Most of the data being generated right now is unstructured, and being able to understand it empowers businesses to uncover hidden patterns; trends; and important information that may be inaccessible through traditional, structured data analysis. Answering the questions in this self-reflection will help reinforce what you've learned and enable you to make the most of all kinds of data throughout your career.

Scenario

Review the following scenario. Then complete the instructions.

In this self-reflection, you’ll explore and reflect on the nature of unstructured data by interacting with a crowd-sourced dataset on the website Quick, Draw!

Quick, Draw! is a game in which people create drawings. Quick! Draw has used these drawings to amass a dataset that contains millions of images separated into categories such as plants, animals, or vehicles. On the Quick, Draw! website, you can examine these drawings or play the game to add your own. These drawings help train computers to recognize objects with artificial intelligence.

In this self-reflection, you’ll explore the Quick! Draw website and create your own drawings. Then, you’ll reflect on Quick! Draw’s dataset and what you can deduce about the nature of unstructured versus structured data.

Step-By-Step Instructions

Follow the instructions to complete each step of the activity. Then answer the questions at the end of the activity before going to the next course item.

Step 1: Explore the Quick, Draw! website

1. Visit the Quick, Draw! website.
2. Click the drop down arrow featured next to **Now visualizing: cloud**.
3. Select a type of doodle to begin.
4. Click an image to get its details, like the number of doodles drawn. For example, there are more than 100,000 different drawings of elephants.
5. Scroll through the list and determine if there are any that don't belong. If you find one that doesn't appear to match the intended object, click on it and select **Flag as inappropriate**.
6. Explore other categories of drawings. Choose three that interest you and examine their doodles.

Optional Step 2: Explore Get the data

Explore further. Click **Get the data** to visit the GitHub page containing the entire dataset. As you become more familiar with data projects and start creating your own, you can return to this dataset and analyze it yourself.

Optional Step 3: Explore Play the game

Click **Play the game** to draw your own doodles and contribute to Quick, Draw!’s dataset.

1.

Question 1

Reflection

Consider the doodles you found in the Quick, Draw! dataset:

- What did you notice as you explored drawings in different categories? Are there consistent themes among the pictures in a category?
- If you didn't know the category labels, how would you distinguish the pictures from one another? What would you look for?

Reflect on your choices and think about categories with the pictures you just explored.

Now, write 2-3 sentences (40-60 words) in response to each of these questions. Type your response in the text box below.

Answer:

There are a huge number of pictures of drawings in each category. In each category some picture is appropriate by its name, and some are different to identify. If the label is not found, then identifying the proper category of the pictures is very difficult. If you want to identify the category, you have to make a guess about seeing most of the similar pictures.

Feedback

Great work reinforcing your learning with a thoughtful self-reflection! An effective response would include how the data (in this case, the drawings) is vastly different across the three categories. In each category, the pictures also vary by a wide margin, almost as much as they do between categories.

Further, within a given category, the pictures may be different, but consistent similarities persist. For example, virtually every picture of an elephant includes a trunk and big ears, while almost every television is rectangular and has an antenna.

2.

Question 2

Consider what you know about structured and unstructured data and how it connects to the Quick, Draw! website:

- How would you describe the Quick, Draw! doodles you explored from a data point of view? For instance, how are these doodles organized? Would you be able to store this type of data in a database?
- How are these doodles different from or similar to other types of data that you have encountered?
- What about this data makes it unstructured?

Reflect on your learning and think about data with the Quick, Draw! doodles you created.

Now, write 2-3 sentences (40-60 words) in response to each of these questions. Type your response in the text box below.

Quick,Draw! Doodles have a huge amount of data, which is unorganized, and in some of the cases, the data will be appropriate. These doodles are organized as an unstructured. Yeah, this is unstructured data; in a database, this type of data is easy to store. In the category, most of the doodles are the same, but some are totally different; we can recognize it by seeing the drawing of the picture. This data isn't organized in rows and columns in the database, so this data is unstructured.

Feedback

Great work reinforcing your learning with a thoughtful self-reflection! An effective reflection on this topic would note that these doodles are unstructured data because they cannot be organized into columns and rows.

For instance, the data on Quick, Draw! is organized loosely based on the category, but not beyond that. Within each category, there is no organization. Unstructured data also has no established rule about how to compare two different pieces of data. On the other hand, structured data conforms to organizational rules.

In the example of the elephant, there are no rules that make one picture more elephant-like than any other. Rules are one way to structure data, as they can act as a test to help determine if a data point (in this case, an image) should or shouldn’t be considered a picture of an elephant.